

Vivian U

Associate Research Scientist · Roman Proposal Lead

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Professional Appointments

Member of the Professional Staff, <i>Caltech</i>	2026–Present
Associate Research Scientist, <i>Caltech / IPAC</i>	2025–Present
Assistant Research Scientist, <i>University of California, Irvine</i>	2020–Present
Assistant Project Scientist, <i>University of California, Irvine</i>	2017–2019
UC Chancellor’s Postdoctoral Fellow, <i>University of California, Riverside / Irvine</i>	2015–2017
Thirty Meter Telescope Postdoctoral Scholar, <i>University of California, Riverside</i>	2012–2014

Education

Ph.D. in Astronomy, <i>University of Hawaii at Manoa (Advisors: D. Sanders & G. Fazio)</i>	2012
B.S. in Astrophysics, <i>California Institute of Technology (Senior Thesis Advisor: R. Ellis)</i>	2006

Selected Awards and Honors

Commended Faculty Mentor UCI Campuswide Honors Collegium	2025
NASA Group Achievement Award (Habitable Worlds Observatory Collaborative Team)	2024
AAAS Newcomb Cleveland Prize	2020
National Academy of Sciences Kavli Frontiers of Science Fellow	2019
University of California Chancellor’s Postdoctoral Fellowship (\$110k)	2015–2017
NASA Harriett G. Jenkins Predoctoral Fellowship (\$114k)	2009–2012
Smithsonian Astrophysical Observatory Predoctoral Fellowship (\$20k)	2009–2012

Selected Telescope and Research Grants

Summary: Served as Principal Investigator (PI), Co-PI, or Institutional PI on 11 selected NASA, NSF, and space telescope programs, securing and managing over \$2.2M in research funding since 2020. Notable awards include NASA ROSES, NSF AAG, JWST, HST, Chandra, and NASA Keck.

NASA Keck 2026B — “From Cosmic Noon to Twilight: KCWI as HWO Precursor” <i>PI (\$35k)</i>	2026–2027
NASA Chandra Cycle 27 — “Confirming dual active intermediate-mass black holes with kiloparsec separation” 2026–2027 <i>Co-I / Admin PI (\$37k; PI: Guo)</i>	
NASA Chandra Cycle 26 — “Confirming a unique triple broad-line and radio-emitting AGN system at kpc-scale” 2026–2027 <i>Co-I / Admin PI (\$14k; PI: Guo)</i>	
JWST Cycle 4 — GO-8391: “A Triad Mystery: Deciphering a Triple BL Radio AGN System at Kpc Separation” 2025–2027 <i>PI (\$100k)</i>	
NASA ADSPS — “Probing Cosmic Ecosystems in the Far-IR: Gas, Dust, and Black Holes across Cosmic Time” 2025–2027 <i>PI (\$666k including subawards)</i>	
NSF AAG — “MAUNA: Multiphase Analysis of (U)LIRG Nuclear Activity” <i>PI (\$461k)</i>	2024–2027

JWST Cycle 1 — GO-1717: “Feedback around Supermassive Black Holes in Dusty Nuclei” <i>PI (\$349k)</i>	2023–2025
NASA ADAP — “Fueling the Fire: Tracing Black Hole Growth in Pre-Quasar Systems” <i>Co-I / Institutional PI (\$168k subaward; PI: Medling)</i>	2023–2026
HST mid-Cycle 30 — “From Galactic Cores to the Cosmic Web – A Study of Feedback and Multiphase Galactic Winds with HST and JWST” <i>PI (\$86k)</i>	2023–2024
HST Cycle 30 — “Calibrating the Black-Hole Mass Scaling Relations using Reverberation-Mapped Active Galaxies with Velocity-Resolved Measurements” <i>Co-I / Institutional PI (\$30k subaward; PI: Bennert)</i>	2023–2024
NASA ADAP — “Hunting for Light in the Dark: A Multiwavelength Guide to Obscured SMBHs” <i>PI (\$279k)</i>	2020–2022

Selected Invited Presentations

Summary: Delivered over 55 Invited Colloquia/Seminars and over 35 Contributed Talks at research institutions, astrophysics centers, and international conferences.

Cosmic Dawn Center (DAWN), Copenhagen, Denmark	Seminar (Jun 2026)
AAS 248 PRIMA Special Session, Pasadena, CA	Featured Talk (Jun 2026)
University of Hawai’i at Manoa, Honolulu, HI	Colloquium (Jul 2025)
George Mason University, Fairfax, VA	Colloquium (Jan 2025)
IAA, Academia Sinica, Taipei, Taiwan	Colloquium (Nov 2024)
Aspen Center for Physics, Aspen, CO	Colloquium (Aug 2024)
STScI / Johns Hopkins University, Baltimore, MD	Joint Colloquium (Apr 2024)
Roman Science Inspired by JWST Results Conference, Baltimore, MD	Featured Talk (Jun 2023)
Olympian Symposium, Paralia Katerini, Mt. Olympus, Greece	Review Talk (May 2023)
IA-FORTH, University of Crete, Heraklion, Crete, Greece	Seminar (May 2023)
University of California San Diego / San Diego State University, virtual	Seminar (Mar 2022)
University of Bath / Bristol / Cardiff, virtual	Seminar (Apr 2021)
Queen’s University Belfast, virtual	Seminar (Apr 2020)
New Mexico State University, Las Cruces, NM	Special Colloquium (Feb 2020)
NRAO / University of Virginia, Charlottesville, VA	Colloquium (Feb 2020)
University of California Santa Barbara, Santa Barbara, CA	Colloquium (Sep 2019)
Texas A&M University, College Station, TX	Seminar (Sep 2019)
National Optical Astronomy Observatory, Tucson, AZ	Special Colloquium (Jun 2019)
University of Melbourne, Victoria, Australia	Special Colloquium (Apr 2019)
Pomona College, Claremont, CA	Colloquium (Feb 2019)
University of Chicago, Chicago, IL	Special Colloquium (Feb 2019)
University of Kansas, Lawrence, KS	Special Colloquium (Feb 2019)
Rochester Institute of Technology, Rochester, NY	Special Colloquium (Feb 2019)
University of California Santa Cruz, Santa Cruz, CA	Special Colloquium (Feb 2018)
Bash Symposium, University of Texas, Austin, Austin, TX	Featured Talk (Oct 2017)
University of Southern California, Los Angeles, CA	Special Colloquium (Aug 2017)
University of California Davis, Davis, CA	Seminar (Mar 2017)
Space Telescope Science Institute, Baltimore, MD	Special Colloquium (Mar 2017)

Selected Professional Leadership and Service

National Infrastructure and Societal Professional Leadership:

- **AAS Nominating Committee Member** (2024–2026) · **AAS Agent** at UC Irvine (2022–2025).
- **Chairperson**, Habitable Worlds Observatory (HWO) Science Interest Group Leadership Council (2025–Present).
- **Co-Chair**, NASA Cosmic Origins AGN Science Interest Group Leadership Council (2022–2026; Co-Chair 2024–2026).
- **Co-Chair**, HWO AGN Working Group (2024–2025) → *Received a Letter of Appreciation from the NASA Science Mission Directorate/Astrophysics Division Director for exceptional leadership and community service (2026)*.
- **Committee Member**, Space Telescope Science Institute (STScI) JWST Users Committee (2025–Present) · IPAC Strategic Conference Science Committee (2025–Present) · IPAC Head of Science Staff Search Committee (2025–2026).

International Governance & Community Panels:

- **Secretary** (2025–Present) and Member (2022–Present), US Liaison Committee for the International Union of Pure and Applied Physics (IUPAP) Commission on Astrophysics (C19).
- **Panel Chair & Reviewer**: NASA ADAP, NSF AAG/AAPF, HST, JWST, Swift, UC Keck Time Allocation Committee (TAC), JPL-Palomar TAC, Canadian TAC, and China’s Telescope Access Program TAC (2017–Present).
- **Conference Organizing**: SOC/LOC for 8 international symposia, including: SOC Chair for ALMA Science Center Workshop (Honolulu, 2020; secured \$7.7k grant); SOC for Evolution of Dust and Gas (Hiroshima, 2024); SOC for Surveying the Universe in 4D (Baltimore, 2026); LOC for NASA Ad ASTRA Workshop (Pasadena, 2026).
- **Journal Referee**: *Nature*, *Astrophysical Journal*, *MNRAS*, *Astronomy & Astrophysics*, *Astr & Space Science*, *SPIE*.

Inclusive Excellence and Campus Climate:

- UCI Committee on Inclusive Excellence / APS-IDEA Team (2017–25) · Advisory Council on Campus Climate, Culture, and Inclusion (2017–20) · Women in Physics Executive Committee (2015–21) · Physical Sciences Climate Council (2022–24).

Mentoring, Outreach, and Teaching

Research Mentoring Profile: Supervised 4 Postdocs, 5 Graduate Students, 10 Undergraduates, and 12 High School Students. Produced 7 mentee-first-author refereed publications.

- *Postdoctoral Fellows*: Yiqing Song (ESO Fellow); Marina Bianchin (IAU Gruber Foundation Fellow / Juan de la Cierva Scholar at IAC); Justin Kader (UCI / IPAC); Benjamin Metha (IPAC)
- *Graduate Students*: Raymond Remigio (PhD co-advised, UCI); Cristina Lofaro (PhD Committee; now postdoc); Tianmu Gao (Masters, now PhD at Australian National University); Shoubaneh Hemmati (co-advised; now Staff Scientist at IPAC); Laura DeGroot (co-advised; now Visiting Assistant Professor, College of Wooster).
- *Undergraduates & High School Mentees*: Hanna Schmidt (Whitman); Max Abrina (UCI); Justin Peterson (CSUB, co-advised); Hajar Aziz (UCR Grad); Judson Leary (UCI); Nikki Azarafza (University High); Liam Hunt (U. of Rochester Grad); Daphne Ghazizadeh (George Washington U, co-advised); Jeffrey Gong (DBHS, co-advised); Akshay Gowrishankar (Caltech SURF, co-advised); Ansh Vashisth (Berkeley); Bailey Liu (Regeneron Top 300 Scholar / Johns Hopkins Undergrad Fellow); Alex Vogler (UCI, co-advised); Thomas Schmidt (UCI, co-advised); Armeen Mobasher (UCR); Alex Herrera (MIT); Michele Dufault (Yale, deceased).

Public Outreach & Scientific Communication Highlights:

- **Global Panels**: Judge for the international *Breakthrough Junior Challenge* (2021–Present).
- **Invited Service Lectures**: NASA Community College Network Shapley Lectures (2025) · National Academies

Board on International Scientific Organizations Global Women’s Breakfast Panelist on Interdisciplinary Inclusivity (2023).

- **Public Engagement:** TEDx Speaker, Manhattan Beach (2024) · Speaker, Lick Observatory Summer Series (2018, 2019) · Speaker, Astronomy on Tap (Aspen, 2025).

Formal Academic Teaching:

- **Guest Lecturer,** UCI Phys 138 (2025) · UCI Osher Lifelong Learning Institute SC 208 (2024)
- **Course Instructor,** University of Hawaii at Manoa Astr 101 (2009) · ISEE / Professional Development Program (2009)

Selected Press Release and Media Coverage

DongA Science: “Science from the Cover: Precessing black hole jet observed for first time in spiral galaxy”	Mar 2026
Universe Today: “Astronomers Discover the First Galaxy-Wide Wobbling Black Hole Jet”	Jan 2026
Scientific American: “Weird, Wobbling Black Hole Jets Can Shape Entire Galaxies”	Jan 2026
NRAO: “NSF VLA Helps Reveal Record-Breaking Gas Stream”	Jan 2026
Keck Observatory & Caltech/IPAC: “Astronomers Discover the First Galaxy-wide Wobbling Black Hole Jet in a Disk Galaxy”	Jan 2026
UCI: “UCI astronomers spot largest known stream of super-heated gas in the universe”	Jan 2026
AAS Press Briefing: “A Precessing Radio Jet Drives Super-Heated Gas Outflow”	Jan 2026
New Scientist: Dead Planets Society: Giving the Milky Way More Arms	May 2024
PBS NOVA: “New Eye on the Universe” Featured Expert Segment	Feb 2023
AAS Press Briefing: “Unveiling the Dusty Hearts of Galaxies with JWST”	Jan 2023
Irvine Weekly: “UCI Astronomers Leading James Webb Space Telescope Research”	Nov 2022
UCI: “UCI-led astronomers capitalize on early access to JWST data”	Nov 2022
New Scientist: JWST has caught two galaxies smashing together and sparking starbursts	Aug 2022
Nature News: “Landmark JWST releases first science image — astronomers are in awe”	Jul 2022
Inverse: “How JWST will unveil the mysteries of cosmic star-making factories”	Jul 2022
UCI: “Galaxies and Supermassive Black Holes, Everyday Life for Vivian U”	Feb 2022
Nature News Feature: “The \$11-billion Webb telescope aims to probe the early Universe”	Dec 2021
California Science Center (podcast): “...how scientists will use JWST? (with Vivian U)”	Dec 2021
Scientific American: “Galaxy Collisions Preview Milky Way’s Fate”	Dec 2021
UCI: “Extragalactic pioneer”	Sep 2020
NASA: “NASA’s Webb Will Explore the Cores of Merging Galaxies”	Sep 2020
Universe Today: “When do Black Holes Become Active? The Case of the Strangely-Shaped Galaxy Mrk 273”	Aug 2013

Publications Summary

Citation Metrics (NASA ADS Profile): H-index: 35 · Total Citations: 4,500+ as of Jul 2026

Refereed Publication Overview: Co-authored a total of 90 refereed papers, including 15 as primary author or primary mentor, 26 as a core co-investigator, and 49 within broader collaborative frameworks.

Refereed Publications (Primary Leadership & Supervised Mentees)

*Work led as PI or first author; *denotes work led by postdocs and students under my primary supervision*

15. *Song, Y.; **U, V.** et al. “The Multi-phase Biconical Outflow in the local IR-Luminous Merger IRASF01364-1042.” 2026, **A&A**, accepted, [arXiv:2606.06005](https://arxiv.org/abs/2606.06005)

14. *Aziz, H.; U, V. et al. “Clumpy Disk, Interloper, or Merger? Nature of a Distant Galaxy Pair at 5 kpc Projected Separation.” 2026, [ApJ](#), 1005, 224
13. *Kader, J.; U, V. et al. “A precessing jet from an active galactic nucleus drives gas outflow from a disk galaxy.” 2026, [Science](#), 391, 911
12. *Remigio, R.; U, V. et al. “Spatially Resolved [O III] Emission Line Kinematics of Reverberation-Mapped AGNs with the Keck Cosmic Web Imager.” 2025, [ApJ](#), 992, 42
11. *Kader, J.; U, V. et al. “Shockingly Effective: Cluster Winds as Engines of Feedback in Starburst Galaxy VV 114.” 2025, [ApJ](#), 988, 230
10. *Gao, T.M.; U, V. et al. “Nuclear Spectral Energy Distributions of Luminous Infrared Galaxies.” 2025, [ApJ](#), 988, 185
9. *Bianchin, M.; U, V. et al. “GOALS-JWST: Gas Dynamics and Excitation in NGC 7469 revealed by NIRSpec.” 2024, [ApJ](#), 965, 103
8. U, V. et al. “GOALS-JWST: Resolving the Circumnuclear Gas Dynamics in NGC 7469 in the Mid-Infrared.” 2022, [ApJL](#), 940, L5
7. U, V. “The Role of AGN in Luminous Infrared Galaxies from the Multiwavelength Perspective.” 2022, [Universe](#), 8, 392 (*Invited Review*)
6. U, V. et al. “The Lick AGN Monitoring Project 2016: Velocity-Resolved $H\beta$ Lags in Luminous Seyfert Galaxies.” 2022, [ApJ](#), 925, 52
5. U, V. et al. “Keck OSIRIS AO LIRG Analysis (KOALA): Feedback in the Nuclei of Luminous Infrared Galaxies.” 2019, [ApJ](#), 871, 166
4. U, V. et al. “A Correlation Between $Ly\alpha$ Spectral Line Profile and Rest-Frame UV Morphology.” 2015, [ApJ](#), 815, 57
3. U, V. et al. “The Inner Kiloparsec of Mrk 273 with Keck Adaptive Optics.” 2013, [ApJ](#), 775, 115
2. U, V. et al. “Spectral Energy Distribution of Local Luminous and Ultraluminous IR Galaxies.” 2012, [ApJS](#), 203, 9
1. U, V. et al. “Distance Determination to M33 with FGLR.” 2009, [ApJ](#), 704, 1120

Refereed Publications (Co-Investigator: Core Team & Major Contributions)

26. Kakkad, D. et al., “GOALS-JWST: Resolved multi-phase molecular gas in IRAS 20551-4250 using JWST and ALMA.” 2026, [MNRAS](#), 549, 781
25. Vidal, E. et al., “Modeling the JWST MIRI Counts, Insights Into the Source Properties and Role of Dust-Obscured AGN.” 2026, [ApJ](#), 1002, 23
24. Agostino, J. et al., “Constraining Nuclear Molecular Gas Content with High-resolution CO Imaging of GOALS Galaxies.” 2026, [ApJ](#), 998, 343
23. Sun, J. et al. “The Intermediate-Mass Black Hole Reverberation Mapping Project: First Detection of Mid-Infrared Lags in Prototypical IMBHs in NGC 4395 and POX 52.” 2025, [ApJL](#), 989, L26
22. Winkel, N. et al., “Unified Black-Hole-Mass-Host-Galaxy Scaling Relations in Active and Quiescent Galaxies.” 2025, [ApJ](#), 978, 115
21. Bohn, T. et al., “GOALS-JWST: The Warm Molecular Outflows of the Merging Starburst Galaxy NGC 3256.” 2024, [ApJ](#), 977, 36
20. Linden, S. et al., “GOALS-JWST: Constraining the Emergence Timescale for Massive Star Clusters in NGC 3256.” 2024, [ApJL](#), 974, L27
19. Zuo, W. et al., “The Intermediate-Mass Black Hole Reverberation Mapping Project: Initial Results for a candidate IMBH in a nearby Seyfert 1 Galaxy.” 2024, [ApJ](#), 974, 288
18. Buiten, V. A. et al. “GOALS-JWST: Mid-Infrared Molecular Gas Excitation Probes the Local Conditions of Nuclear Star Clusters and the AGN in the LIRG VV 114.” 2024, [ApJ](#), 966, 166
17. Lai, T. S.-Y. et al. “GOALS-JWST: Small neutral grains and enhanced 3.3 micron PAH emission in the Seyfert galaxy NGC 7469.” 2023, [ApJL](#), 957, L26
16. Linden, S. et al. “GOALS-JWST: Revealing the Buried Star Clusters in the Luminous Infrared Galaxy VV 114.” 2023, [ApJL](#), 944, L55
15. Rich, J. et al. “GOALS-JWST: Pulling Back the Curtain on the AGN and Star Formation in VV 114.”

- 2023, [ApJL](#), **944**, L50
14. Bohn, T. et al. “GOALS-JWST: NIRCам and MIRI Imaging of the Circumnuclear Starburst Ring in NGC 7469.” 2023, [ApJL](#), **942**, L36
 13. Armus, L.; Lai, T.; U, V. et al. “GOALS-JWST: Mid-Infrared Spectroscopy of the Nucleus of NGC 7469.” 2023, [ApJL](#), **942**, L37
 12. Lai, T.; Armus, L.; U, V. et al. “GOALS-JWST: Tracing AGN Feedback on the Star-Forming ISM in NGC 7469.” 2022, [ApJL](#), **941**, L36
 11. Evans, A. et al. “GOALS-JWST: Hidden Star Formation and Extended PAH Emission in the Luminous Infrared Galaxy VV 114.” 2022, [ApJL](#), **940**, L8
 10. Inami, H. et al. “GOALS-JWST: Unveiling the Heavily Dust Obscured Compact Sources in the Merging Galaxy IIZw096.” 2022, [ApJL](#), **940**, L6
 9. Montano, J. et al. “Optical Continuum Reverberation in Dwarf Seyfert Nucleus of NGC 4395.” 2022, [ApJL](#), **934**, L37
 8. Villafañá, L. et al. “The Lick AGN Monitoring Project 2016: Dynamical Modeling of Velocity-Resolved H β Lags in Luminous Seyfert Galaxies.” 2022, [ApJ](#), **930**, 52
 7. Liu, W. et al. “Elliptical Galaxy in the Making: The Dual Active Galactic Nuclei and Metal-enriched Halo of Mrk 273.” 2019, [ApJ](#), **872**, 39
 6. Iwasawa, K.; U, V. et al. “Testing a Double AGN Hypothesis for Mrk 273.” 2018, [A&A](#), **611**, 71
 5. Davies, R.; Medling, A.; U, V. et al. “Reconstructing merger timelines using star cluster age distributions: the case of MCG+08-11-002.” 2016, [MNRAS](#), **458**, 158
 4. Medling, A.; U, V. et al. “Following Black Hole Scaling Relations Through Gas-Rich Mergers.” 2015, [ApJ](#), **803**, 61
 3. Medling, A.; U, V. et al. “Shocked gas in IRAS F17207-0014: ISM collisions and outflows.” 2015, [MNRAS](#), **448**, 2301
 2. Medling, A.; U, V. et al. “Stellar and Gaseous Nuclear Disks Observed in Nearby (U)LIRGs.” 2014, [ApJ](#), **784**, 70
 1. Armus, L. et al. “GOALS: The Great Observatories All-sky LIRG Survey.” 2009, [PASP](#), **121**, 559

Refereed Publications (Other Collaboration Co-Authorships)

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| 49. Agostino, J. et al. 2026, ApJ , accepted, arXiv:2606.19285 | 29. Ricci, C. et al. 2021, MNRAS , 506 , 5935 |
| 48. Villafañá, L. et al. 2026, ApJ , 1004 , 151 | 28. Song, Y. et al. 2021, ApJ , 916 , 73 |
| 47. Riffel, R. et al. 2026, A&A , 710 , 231 | 27. Liu, W. et al. 2020, ApJ , 905 , 166 |
| 46. Aravindan, A. et al. 2026, ApJ , 1000 , 230 | 26. Kool, E. et al. 2020, MNRAS , 498 , 2167 |
| 45. Bennert, V. et al. 2026, ApJ , 1000 , 48 | 25. Treister, E. et al. 2020, ApJ , 890 , 149 |
| 44. Johnstone, M. et al. 2025, ApJ , 995 , 108 | 24. Larson, K. L. et al. 2020, ApJ , 888 , 92 |
| 43. Riesco, C. et al. 2025, ApJ , 990 , 136 | 23. Fischer, T. et al. 2019, ApJ , 887 , 200 |
| 42. Donnelly, G. P. et al. 2025, ApJ , 983 , 79 | 22. Medling, A. M. et al. 2019, ApJL , 885 , L21 |
| 41. Wang, S. et al. 2025, ApJ , 983 , 45 | 21. Herrero-Illana, R. et al. 2019, A&A , 628 , 71 |
| 40. Saravia, A. et al. 2025, ApJ , 979 , 217 | 20. Bannister, K. W. et al. 2019, Science , 365 , 565 |
| 39. Alatalo, K. et al. 2024, ApJ , 975 , 241 | 19. Linden, S. T. et al. 2019, ApJ , 881 , 70 |
| 38. Villafañá, L. et al. 2024, ApJ , 966 , 106 | 18. Torres-Albà, N. et al. 2018, A&A , 620 , 140 |
| 37. Woo, J.-H. et al. 2024, ApJ , 962 , 67 | 17. Treister, E. et al. 2018, ApJ , 854 , 83 |
| 36. Cho, H. et al. 2023, ApJ , 953 , 142 | 16. Riess, A. G. et al. 2018, ApJ , 853 , 126 |
| 35. Aravindan, A. et al. 2023, ApJ , 950 , 33 | 15. Larson, K. L. et al. 2016, ApJ , 825 , 128 |
| 34. Villafañá, L. et al. 2023, ApJ , 948 , 95 | 14. Stierwalt, S. et al. 2014, ApJ , 790 , 124 |
| 33. Song, Y. et al. 2022, ApJ , 940 , 52 | 13. Rodney, S. et al. 2014, AJ , 148 , 13 |
| 32. Guo, H. et al. 2022, ApJ , 927 , 60 | 12. Graur, O. et al. 2014, ApJ , 783 , 28 |
| 31. Linden, S. et al. 2021, ApJ , 923 , 278 | 11. Wang, J. et al. 2014, ApJ , 785 , 55 |
| 30. Cho, H. et al. 2021, ApJ , 921 , 98 | 10. Mazzarella, J. et al. 2012, AJ , 144 , 125 |
| | 9. Iwasawa, K. et al. 2011, A&A , 529 , 106 |

8. Iwasawa, K. et al. 2011, [A&A](#), 528, 137
7. Petric, A. et al. 2011, [ApJ](#), 730, 28
6. Haan, S. et al. 2011, [AJ](#), 141, 100
5. Inami, H. et al. 2010, [AJ](#), 140, 63
4. Howell, J. et al. 2010, [ApJ](#), 715, 572
3. Aspin, C. et al. 2009, [ApJ](#), 692, 67
2. MacArthur, L. A. et al. 2008, [ApJ](#), 680, 70
1. Conselice, C. J. et al. 2008, [MNRAS](#), 383, 1366

Selected Non-Refereed Publications, White Papers, & Technical Reports

8. U, V. et al. “Unveiling Intermediate-Mass Black Holes through Reverberation Mapping with the Habitable Worlds Observatory.” 2026, [ASPC](#), 543, 79
7. U, V. et al. “Total Eclipse of the Heart of Galaxies with Roman CGI.” 2025, Roman Community Participation Program
6. Vaccari, L. et al. “Sustainability & Inclusivity: An Interdisciplinary Conversation.” 2023, [Chemistry International](#), 45, 44
5. Koss, M.; U, V. et al. “BH Growth in Mergers and Dual AGN.” 2019, Astro2020 Decadal Survey, [BAAS](#), 51, 504
4. Kelley, L. Z. et al. “Multi-Messenger Astrophysics with Pulsar Timing Arrays.” 2019, Astro2020 Decadal Survey, [BAAS](#), 51, 490
3. Gültekin, K. et al. “Black Holes Across Cosmic Time.” 2019, Astro2020 Decadal Survey, [BAAS](#), 51, 287
2. Lopez-Rodriguez, E. et al. “Tracing the Feeding and Feedback of Active Galaxies.” 2019, Astro2020 Decadal Survey, [BAAS](#), 51, 138
1. Skidmore, W. et al. “Thirty Meter Telescope Detailed Science Case: 2015.” 2015, [RAA](#), 15, 1945